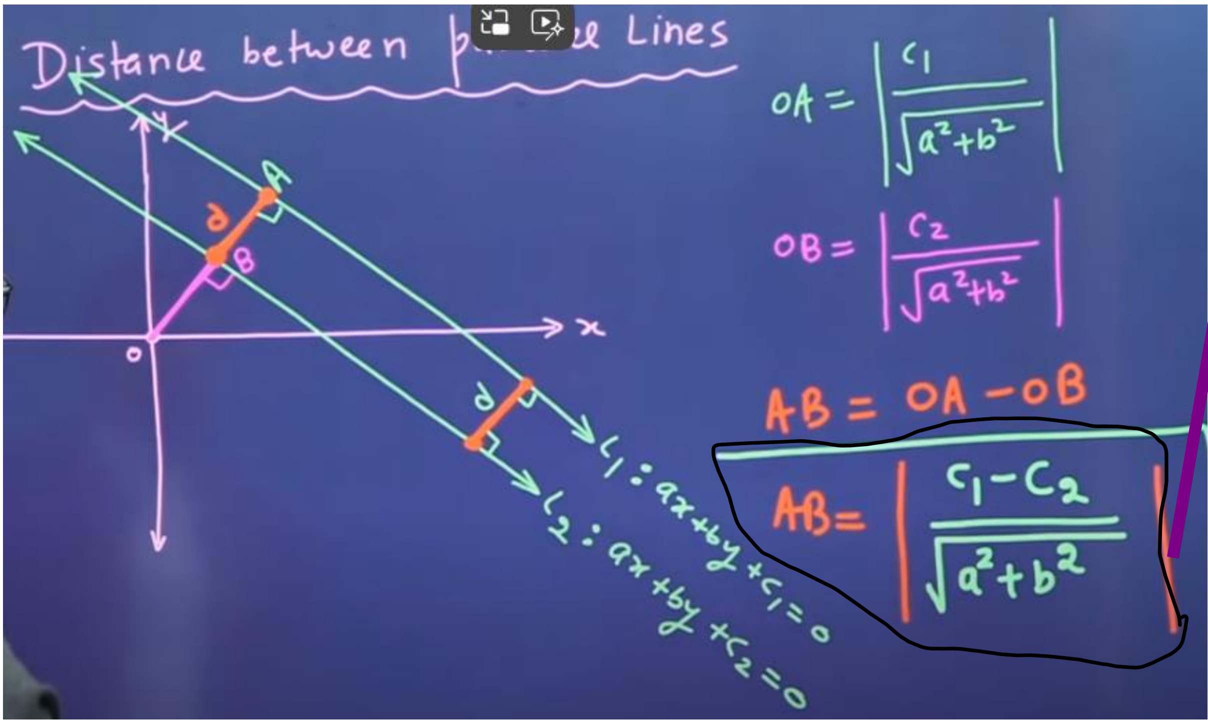
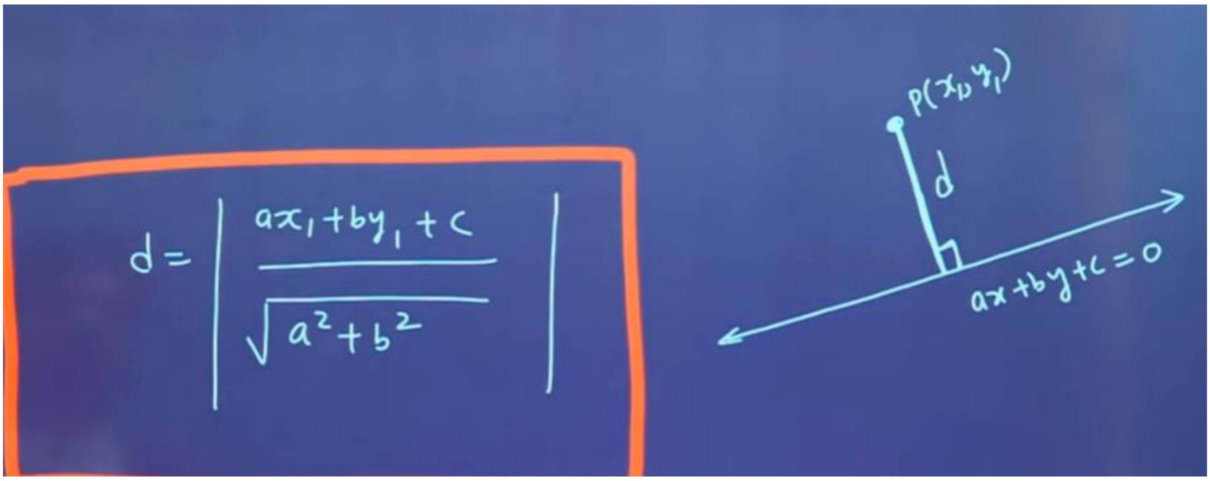
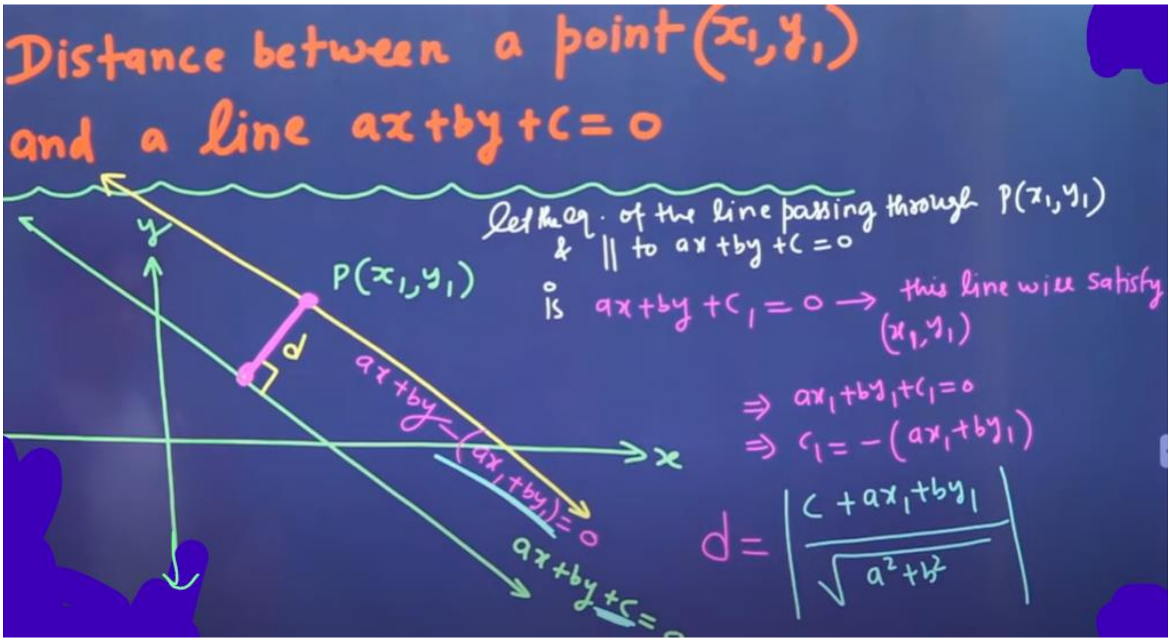
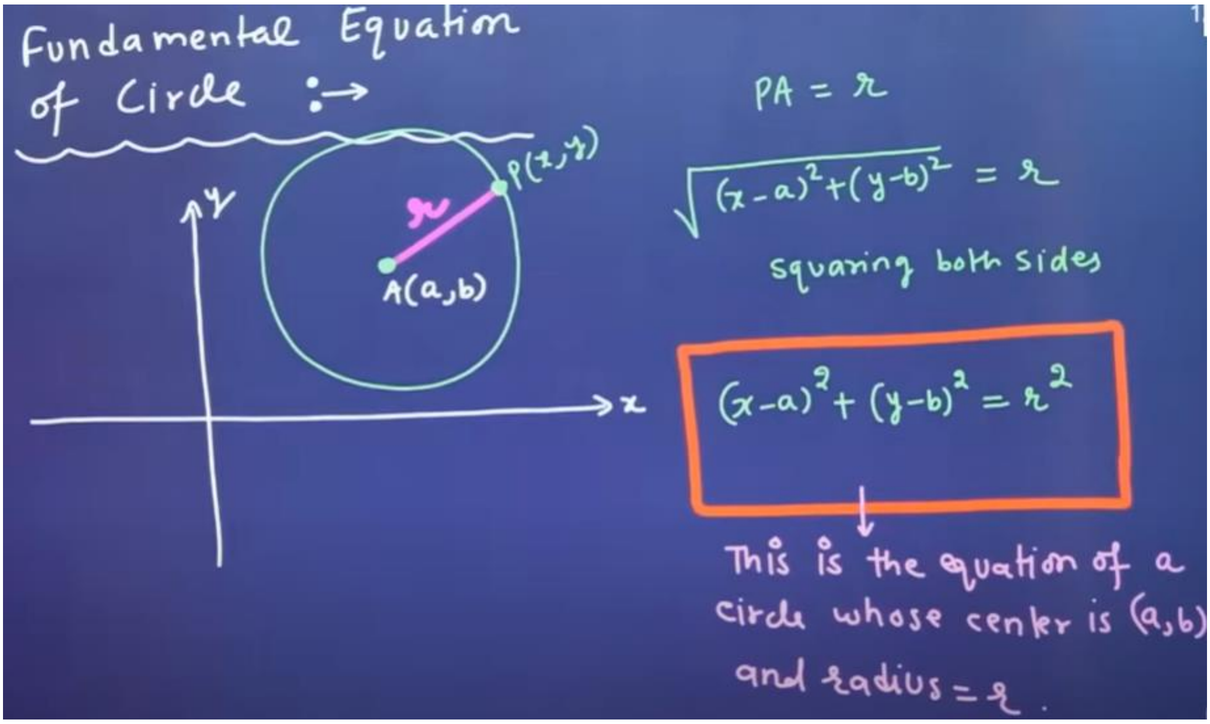


- C1 and C2 must be on same side
- Coeff. of X & Y must be same in both lines




Distance between a point (x_1, y_1)
and a line $ax+by+c=0$
Let the eq. of the line passing through $P(x_1, y_1)$
& $\parallel$ to $ax+by+c=0$
is $ax+by+c_1=0 \rightarrow$ this line will satisfy (x_1, y_1)
$\Rightarrow ax_1+by_1+c_1=0$
$\Rightarrow c_1 = -(ax_1+by_1)$
$d = \left|\frac{c+ax_1+by_1}{\sqrt{a^2+b^2}}\right|$
$P(x_1, y_1)$
d
$ax+by-(ax_1+by_1)=0$
$ax+by+c=0$
y
x


Fundamental Equation of Circle :→
$PA = r$
$\sqrt{(x-a)^2+(y-b)^2} = r$
squaring both sides
$(x-a)^2+(y-b)^2 = r^2$
This is the equation of a circle whose center is (a,b) and radius = r.
P(x,y)
r
A(a,b)
y
x

General Equation of Circle

$$(x-a)^2+(y-b)^2=r^2$$

center $\equiv (a,b)$

radius $= r$

$$x^2+a^2-2ax+y^2+b^2-2by=r^2$$

$$x^2+y^2-2ax-2by+\underbrace{(a^2+b^2-r^2)}_{c}=0$$

put $-a=g$

$-b=f$

$$\boxed{x^2+y^2+2gx+2fy+c=0}$$

General Eq. of circle

center $\equiv (-g,-f)$

$\left(-\frac{1}{2}\text{coeff. of } x, -\frac{1}{2}\text{coeff. of } y\right)$

Radius $=\sqrt{g^2+f^2-c}$

Note:- $r=\sqrt{g^2+f^2-c}$

Point → A circle of Radius zero.

(i) If $g^2+f^2-c>0$ $\Rightarrow$ Then circle is real

(ii) If $g^2+f^2-c=0$ $\Rightarrow$ Then it is a point circle.

(iii) If $g^2+f^2-c<0$ $\Rightarrow$ Then it will an imaginary circle.

$x^2+y^2+x+y+4=0$ → This is an imaginary circle.